

YANG LIU

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EDUCATION

- Ph.D. in Computer Science, Virginia Tech 2025 – 2030 (expected)
- M.S. in Software Engineering, Xi'an Jiaotong University 2022 – 2025
- B.S. in Software Engineering, Northwest A&F University 2018 – 2022

RESEARCH PROJECTS

Conflict-Aware Agent Memory

- Designed SoG (first author, in submission): a training-free sensemaking layer that resolves conflicting facts in agent memory via provenance-aware claim graphs and a gated resolution ladder (temporal recency, Bayesian source reliability, LLM-judge audit); reaches **95.0%/54.0% single/multi-hop accuracy** on MemoryAgent-Bench FactConsolidation vs. 76%/13% for the strongest RAG baseline, consistent across three backbones (GPT-4o-mini, Qwen2.5-7B/72B).
- Outperformed the multi-agent-debate SOTA (MADAM-RAG, COLM '25, official code) on its native RAMDocs benchmark (17.4 vs. 16.6 exact set-EM) at **$\sim 1/7$ the inference cost** (2.6 vs. 18.6 LLM calls/query), and swept the temporal HoH benchmark (64.3% vs. $\leq 1.7\%$ for six prompting-control baselines).
- Built the end-to-end evaluation harness: dataset-agnostic pipeline with a gold-label firewall and bit-exact replayable inference caches; ran **8 methods $\times$ 3 backbones $\times$ 5 conflict benchmarks** on SLURM A100/H200 clusters (vLLM-served open models), auditing and invalidating two leakage-contaminated legacy results.

LLM-Grounded Interactive Visualization

- Built CriterionSI (first author, IEEE VIS '26): an LLM-powered criterion tracker that infers users' evolving clustering criterion from sequential drag interactions, propagated to the layout through criterion-guided re-projection (LLM-assigned semantic constraints, drag constraints, incremental MDS); commits to the correct criterion within $\sim 10\%$ of dataset-size interactions and improves clustering quality **69% over the strongest interactive baseline** (silhouette 0.44 vs. 0.26).
- Implemented the full prototype—React front end and Python/LLM pipeline (EVA-CLIP, Gemini 2.5 Flash)—using agentic coding workflows (Claude Code, Codex); also contributed to SVS (IEEE VIS '26), a visual analytics system for verifying multi-document summaries via spatial layouts and coordinated provenance visualization.

LLM Evaluation and Benchmarking

- Co-built CTIConnect (KDD '26, Datasets & Benchmarks Track): a benchmark of **1,860 expert-verified QA pairs** across 9 tasks, integrating 5 heterogeneous CTI sources (CVE, CWE, CAPEC, MITRE ATT&CK, and unstructured threat reports).
- Deployed and evaluated **10 state-of-the-art LLMs** (open-weight models served via vLLM; commercial APIs); built the end-to-end evaluation pipeline comparing retrieval-augmentation strategies (retrieve-then-rerank, IR-CoT, and domain-specific retrieval), revealing task-dependent retrieval bottlenecks.

Deep Learning for Scientific Imaging

- First-authored HSiDMamba (arXiv, 29 citations): among the first bidirectional state-space models for hyperspectral denoising, with linear-complexity spatial-spectral modeling that outperforms transformer baselines.
- First-authored a spectral aggregation cross-square transformer for hyperspectral denoising (ICPR '24); co-authored region-aware sequence-to-sequence learning (ECCV '24) and Fourier-spatial-spectral bridging (ACM MM '24), owning model design, experiments, figures, and writing across the series.

- Earlier, built a binocular-camera trajectory-prediction pipeline for drone tracking (EPnP, detection, tracking) and SiamMask-based tracking software for rumen protozoa video analysis.

SELECTED PUBLICATIONS

1. [Drag, Infer, Reproject: Grounding LLMs through Spatial Interaction for Image Clustering](#)
Yang Liu, Xuxin Tang, Jiahao Xu, Chris North. *IEEE VIS (Short Papers)*, 2026.
2. [Spatial Visual Analytics for Multi-Document Summary Verification](#)
Jiahao Xu, Wei Liu, [Yang Liu](#), Eric Krokos, Kirsten Whitley, Xuan Wang, Rebecca Faust, Chris North. *IEEE VIS (Full Papers)*, 2026.
3. [CTIConnect: A Benchmark for Retrieval-Augmented LLMs over Heterogeneous Cyber Threat Intelligence](#)
Yutong Cheng, [Yang Liu](#), Changze Li, Dawn Song, Peng Gao. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (Datasets & Benchmarks Track)*, 2026.
4. [HSIDMamba: Exploring Bidirectional State-Space Models for Hyperspectral Denoising](#)
[Yang Liu](#), JiaHua Xiao, Xiang Song, Yu Guo, Peilin Jiang, Haiwei Yang, Fei Wang. *CoRR arXiv*, 2024 (29 citations).
5. Region-Aware Sequence-to-Sequence Learning for Hyperspectral Denoising
JiaHua Xiao, [Yang Liu](#), Xing Wei. *European Conference on Computer Vision (ECCV)*, 2024.
6. Bridging Fourier and Spatial-Spectral Domains for Hyperspectral Image Denoising
JiaHua Xiao, [Yang Liu](#), Shuhang Zhang, Xing Wei. *ACM International Conference on Multimedia (ACM MM)*, 2024.
7. Spectral Aggregation Cross-Square Transformer for Hyperspectral Image Denoising
[Yang Liu](#), Yantao Ji, JiaHua Xiao, Yu Guo, Peilin Jiang, Haiwei Yang, Fei Wang. *International Conference on Pattern Recognition (ICPR)*, 2024.

OPEN-SOURCE SOFTWARE

- [CriterionSI](#) — code and data for the IEEE VIS '26 paper: LLM criterion tracker, criterion-guided reprojection, and simulation-based evaluation harness.
- [RAS2S](#) — official PyTorch implementation of the ECCV '24 region-aware sequence-to-sequence hyperspectral denoising paper.
- [HSID](#) — general codebase for hyperspectral denoising experiments (training, evaluation, baselines).

TECHNICAL SKILLS

LLM: OpenAI / Gemini / Claude APIs, prompt engineering & structured outputs, agent memory, RAG pipelines, LLM-as-judge & benchmark design, vLLM serving, Hugging Face Transformers

ML & Infrastructure: PyTorch (multi-GPU training), Transformer & state-space architectures, CUDA (kernel-level modification), large-scale evaluation harnesses on SLURM (A100/H200) clusters

Programming & Tools: Python, C++, MATLAB; React prototyping via agentic coding (Claude Code, Codex); Linux, Git

HONORS AND AWARDS

- National Scholarship of China (top 1%), Xi'an Jiaotong University 2023, 2024
- Outstanding Graduate Student (top 5%), Xi'an Jiaotong University 2023, 2024
- First-Class Scholarship, Shenzhen Stock Exchange 2023